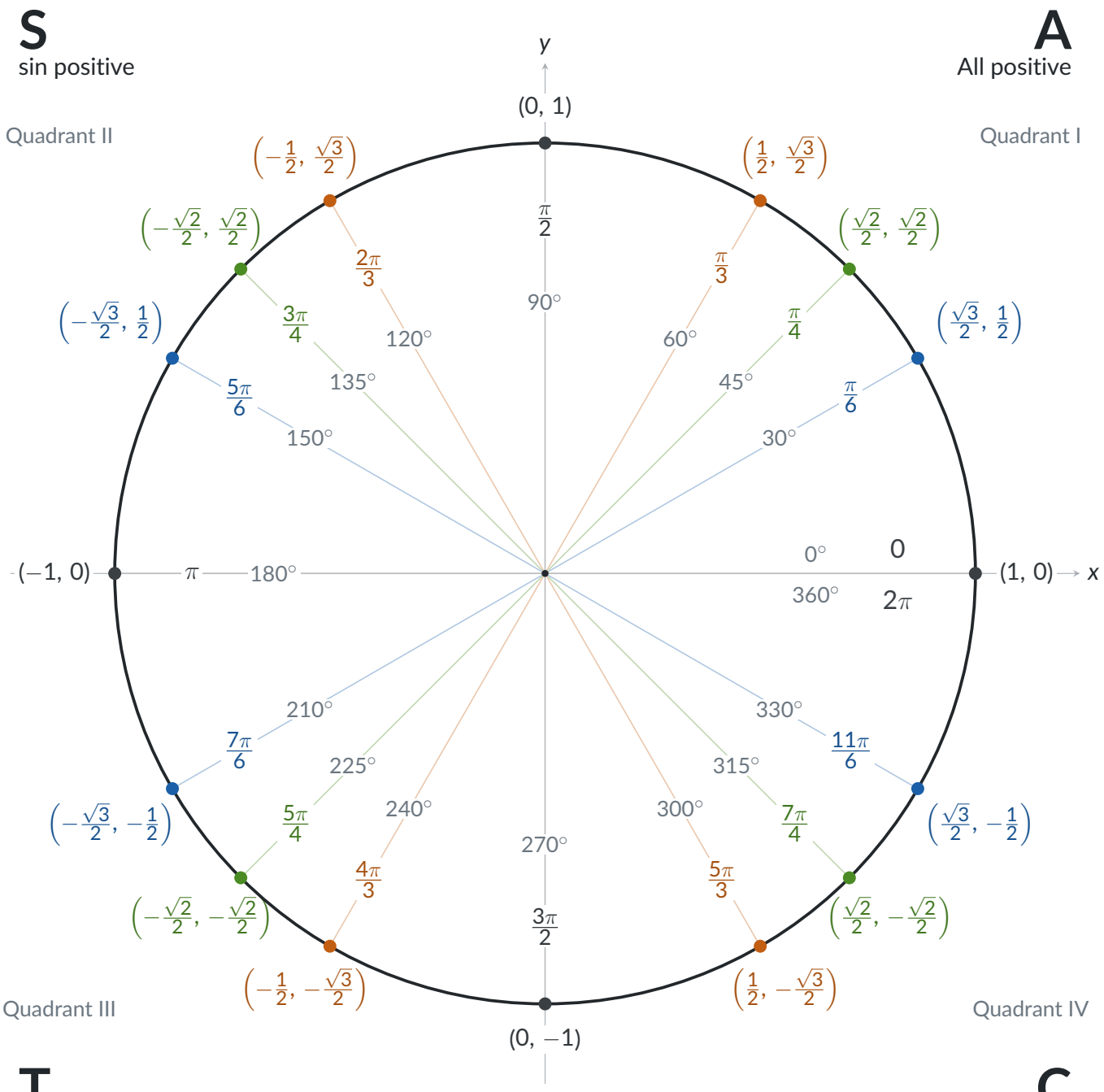
The Unit Circle

Radius 1. Every point is $(\cos\theta,\ \sin\theta) = (x,\ y)$.

Name _____



• denominator 6

reference angle $\frac{\pi}{6} = 30°$

values $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$

• denominator 4

reference angle $\frac{\pi}{4} = 45°$

values $\frac{\sqrt{2}}{2}$ and $\frac{\sqrt{2}}{2}$

• denominator 3

reference angle $\frac{\pi}{3} = 60°$

values $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$

• on an axis

values are $0, \pm 1$

Unit Circle Values

$\sin\theta = y$, $\cos\theta = x$, $\tan\theta = \dfrac{\sin\theta}{\cos\theta} = \dfrac{y}{x}$ (the slope of the radius)

deg	rad	$\sin\theta$	$\cos\theta$	$\tan\theta$
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	undefined
120°	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
135°	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
150°	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
180°	π	0	-1	0
210°	$\frac{7\pi}{6}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
225°	$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	1
240°	$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\sqrt{3}$
270°	$\frac{3\pi}{2}$	-1	0	undefined
300°	$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-\sqrt{3}$
315°	$\frac{7\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-1
330°	$\frac{11\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$

The square-root ladder

Learn one line. Everything else is signs.

θ	0°	30°	45°	60°	90°
sin	$\frac{\sqrt{0}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{4}}{2}$
cos	$\frac{\sqrt{4}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{0}}{2}$

sin counts up 0, 1, 2, 3, 4. cos counts down.

The hand trick

Left hand, palm toward you, fingers spread. Pinky = 0°, ring = 30°, middle = 45°, index = 60°, thumb = 90°.

Fold down the finger for your angle.

$\sin\theta = \dfrac{\sqrt{p}}{2}$, p = fingers on the pinky side

$\cos\theta = \dfrac{\sqrt{t}}{2}$, t = fingers on the thumb side

Example: fold the ring finger (30°). 1 finger on the pinky side, 3 on the thumb side: $\sin 30° = \frac{\sqrt{1}}{2} = \frac{1}{2}$, $\cos 30° = \frac{\sqrt{3}}{2}$.

Big or small?

Near the x-axis, y is small.
30°: $\sin = \frac{1}{2}$, $\cos = \frac{\sqrt{3}}{2}$

Near the y-axis, y is big.
60°: $\sin = \frac{\sqrt{3}}{2}$, $\cos = \frac{1}{2}$

Check: $\frac{\sqrt{3}}{2} \approx 0.87$ and $\frac{\sqrt{2}}{2} \approx 0.71$.

Any angle in three steps

1. Family. Look at the denominator: 6, 4 or 3 gives reference angle $\frac{\pi}{6}$, $\frac{\pi}{4}$ or $\frac{\pi}{3}$.
2. Size. Read the value off the ladder.
3. Sign. Find the quadrant. Use **A**ll **S**tudents **T**ake **C**alculus.

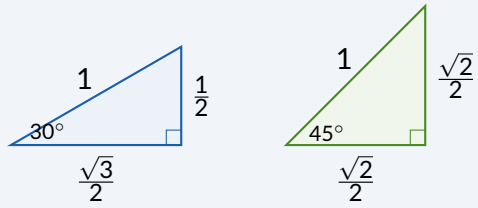
Example: $\sin\dfrac{7\pi}{6}$

denominator 6 → size $\frac{1}{2}$

$\frac{7\pi}{6}$ is just past π → Quadrant III

sin is negative there → $\boxed{-\frac{1}{2}}$

Where the numbers come from



The radius is the hypotenuse, so it is 1. The legs are $\cos\theta$ and $\sin\theta$.

Worth knowing

Degrees → radians	$\times\,\pi/180°$
Radians → degrees	$\times\,180°/\pi$
1 radian	$\approx 57.3°$
Pythagoras	$\sin^2\theta + \cos^2\theta = 1$
Same point	$\theta \pm 2\pi$ (coterminal)
Negative angles	$\cos(-\theta) = \cos\theta$
	$\sin(-\theta) = -\sin\theta$
Order	(cos, sin): c before s, x before y